

CYTISINICLINE

Approval Timeline and Status



FDA Review Delay: The U.S. Food and Drug Administration (FDA) issued a Complete Response Letter (CRL) in June 2026, deferring approval due to general manufacturing facility issues from a third-party partner rather than any safety or effectiveness concerns



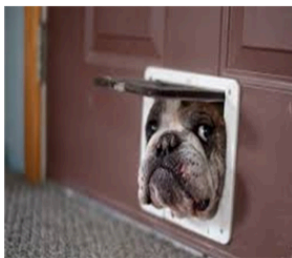
Resubmission Plans: Achieve Life Sciences plans to resubmit its New Drug Application in the fourth quarter of 2026 using a new primary manufacturing partner, Adare Pharma Solutions



Expected Availability: If the resubmission goes well, approval and a subsequent U.S. commercial launch could happen during the first half of 2027

Greatest Success Strategy

One size does
not fit all



Stead et al.
(2016)

Medications can be adjusted

A second agent can be added if there is partial response to quitting smoking.

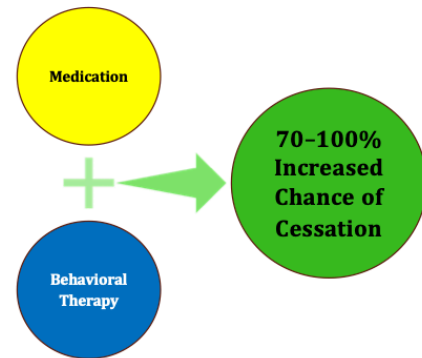
Clinical benefit BUT risk of developing possible side effects to both medications

Examples:

Chantix & NRT

Zyban & NRT

Chantix & Zyban



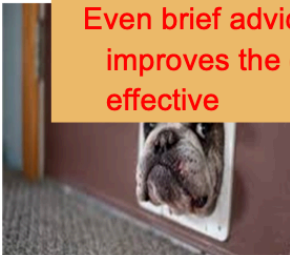
Greatest Success Strategy

One size does
not fit all

Medications can be adjusted

A second agent can be added if there
is partial response to quitting

Even brief advice (less than 3 minutes) from a health professional
improves the chance of quitting successfully and is highly cost-
effective



Stead et al.
(2016)

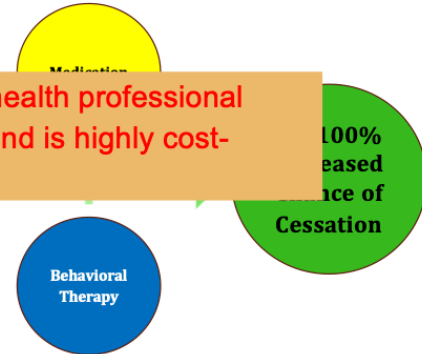
both medications

Examples:

Chantix & NRT

Zyban & NRT

Chantix & Zyban



Cannabis



Plant-derived psychoactive substance containing active chemical compounds called cannabinoids, most notably THC (which alters mood and causes a "high") and CBD (which is non-psychoactive)

THC binds to cannabinoid receptors

These receptors regulate mood, appetite, memory, and pain



Cannabis can be consumed through smoking, vaping, eating, or dabbing, in which people heat cannabis concentrates and inhale the vapor.

Inhalation has the fastest absorption and onset of effect is almost immediate, peaking within 15 min

Cannabis



Marijuana is the term used for consuming the part of the cannabis plant to get “high”



Marijuana Use is Increasing

15% of Americans are active users

22% aged 12+ have used within the last year
(64.2M people)

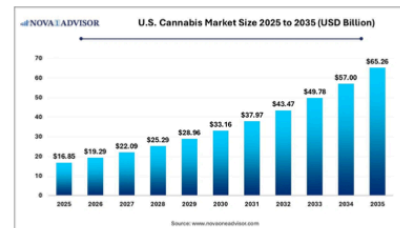
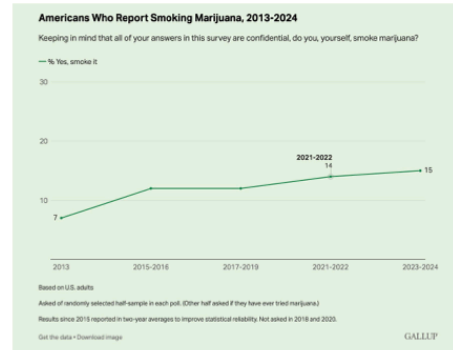
7% of adults > 65 used w/in the last month

Overall, use among those 65 and older surged
nearly 46% from 2021 to 2023

More than 1 in 3 women consume cannabis in
the U.S.

Americans are choosing this over alcohol and
prescription drugs for anxiety, sleep, and
pain management

Cannabis sales boost the economy



Marijuana Use in Teens: stable since 2021

2025 Data (Monitoring the Future Survey)

8th Graders: ~8%

10th Graders: ~16%

12th Graders: ~26%

Teens prefer using the Vapor Pen



Inhaling Marijuana: Is there a 'healthier way'?

Comparison of Health Risks

Risk Category	Smoking Marijuana	Vaping Marijuana
Combustion Toxins	High; releases tar, carbon monoxide, and carcinogens.	Low; heats the element below combustion levels.
THC Potency	Moderate; usually caps around 15% to 25% THC.	Extreme; concentrates can reach up to 95% THC.
Lung Injury Risk	Chronic; causes bronchitis, chronic cough, and phlegm.	Acute & Severe; linked to EVALI and lung chemical burns.
Chemical Additives	None (if smoking pure flower).	High; risks from heavy metals, thinning agents, and pesticides.



Just a DAB of marijuana: 'healthier'?

Traditional smoking of flower creates combustion byproducts and carcinogens

Dabs—concentrated forms of cannabis

Thought to be easier on the lungs by consumers –less smoke and eliminates the byproducts and carcinogens of regular flower smoking

Dabs are either solventless or solvent-based. Solventless do not use butane, propane, or alcohol during extraction. Other Dabs can have such solvent residual.

There are many types of Dabs on the market



Dabbing Marijuana: Not so 'healthy'!

Potency Differences

Regular Flower: Contains about 5% to 25% THC

Dabs (Concentrates): Contain 50% to 90% THC

The Impact: Dabs hit your brain much faster and harder, leading to a higher risk of panic, paranoia, and temporary psychosis



Lung and Chemical Risks



Smoking: Produces tar and plant carcinogens from burning regular cannabis

Dabbing: Uses extreme heat to vaporize a solid or liquid extract

The Impact: While you avoid plant matter, high-heat nails can release toxic byproducts and carcinogens like benzene. Poorly made extracts may also contain leftover butane or chemical solvents.

Dabbing Marijuana: Not so 'healthy'!

Addiction and Tolerance

Because dabs deliver a massive dose of THC all at once, your body builds a tolerance much faster.

Faster tolerance leads to a higher chance of dependency, intense cravings, and difficult withdrawal symptoms.



Dabbing Marijuana: Increased Lung Risk

Chronic lung damage from repetitive
high temperature vapor injury

- cough, wheeze, dyspnea

- obstructive lung disease

Acute hypersensitivity pneumonitis

Organizing pneumonia

Acute respiratory failure



Short-term effects related to brain function may include:



Dizziness.



Mood changes.



Impaired memory.



Disorientation.



Problem-solving issues.



Sleepiness.

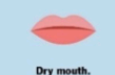


Coordination issues.



Altered senses.

Common short-term effects may include:



Dry mouth.



Nausea.



Increased heart rate.



Increase in appetite.

Health Risks of Marijuana

Underage Marijuana Use Increases the Risk for:



Marijuana Dependence



Worsened Mental Health



Dangerous Accidents



Impaired Brain Development



Poor Academic Performance



Cannabis Use Disorder

Uses in larger amounts than needed

Cravings

Problems at work, school, or home due to use

Spends lots of time on cannabis use

Difficulty controlling or cutting down use

Needs more or more potent cannabis to get the same effect

Keeps using despite social or relationship problems

Stops doing other activities enjoyed to use cannabis

Uses in high-risk situations (like while driving)

Continues use despite physical or psychological problems

Not using causes: irritability, sleep or appetite problems, nausea, abdominal pain

LONG-TERM EFFECTS

- RESPIRATORY PROBLEMS
- DEPRESSION
- LOWERED IMMUNE SYSTEM
- HIGHER RISK FOR ABUSE OF OTHER DRUGS
- CANCER

Case Studies

Case 1

23-year-old college student, previously healthy, presents with fever for 4 days, associated body aches, fatigue and non-productive cough.

ACUTE INFLUENZA INFECTION

~ weekly use of vaping
for Marijuana

Case 2

58-year-old, previous smoker, stopped 10 years ago, presents with three weeks of cough with scarce yellowish sputum, severe exertional dyspnea and weakness.

IDIOPATHIC PULMONARY FIBROSIS

Previous Nicotine
Smoker, stopped 10 years
prior

Case 3

13-year-old male, known history of severe eczema and environmental allergies, presents with progressive exertional dyspnea and wheeze, improves with rest and nocturnal cough.

PERSISTENT POORLY CONTROLLED ASTHMA

Denies smoking, Father
smokes in home ~ ½ PPD

Review Current and Past Smoking Practices with all of your patients! Modifiable Risk Factor!

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



Technique & Interpretation of the lung and thorax PE



Basic pulmonary diagnostic techniques



Understand High Risk behavior for lung disease



Review Vaccine Preventable Respiratory Illness

Review additional Preventable Diseases



PREVENTION

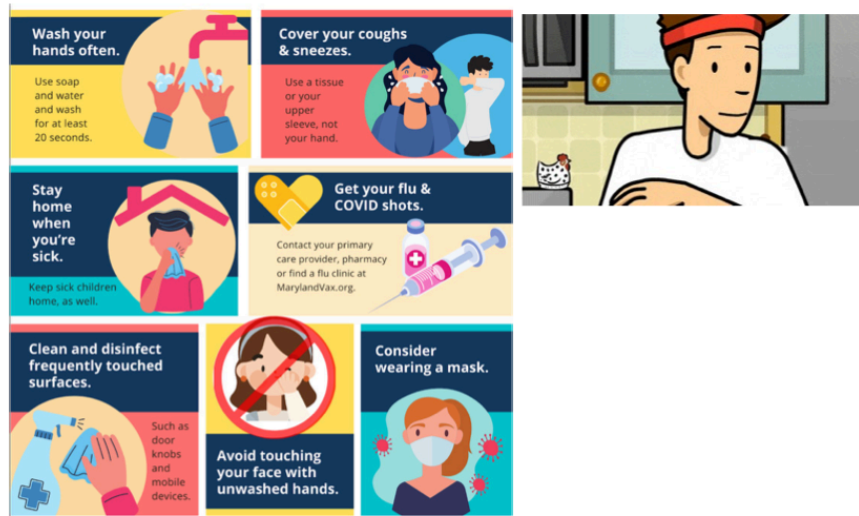
Vaccine Preventable Respiratory Disease

Influenza
COVID-19
RSV
Pertussis
Pneumococcus



Protection Measures for Respiratory Illness

Maryland Department of Health



EPIDEMIOLOGY OF INFLUENZA

Annual seasonal influenza epidemics □
290,000–650,000 deaths worldwide and
infect up to 20% of the population

Approximately 90% of influenza deaths
are seen in those >65yo

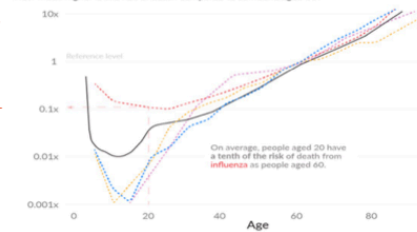
Infant deaths are mainly: Age <5 years
(particularly <2 years, and among those
<2 years, particularly those <6 months)

The risk of death from respiratory diseases
increases exponentially with age

Infants and the elderly tend to have heightened risks of death.

Relative risk of death

How much higher is the risk of death compared to someone aged 60?



Mortality data in this study was sourced from different countries: influenza (US), COVID-19 (France),
SARS (China), MERS (Saudi Arabia), all cases (US European countries).
Data sources: C. Simonsen et al. (2012). *Emerging Infectious Diseases*.
OurWorldInData.org - Research and data to make progress against the world's largest problems.



INFLUENZA PATHOGENESIS & PATHOPHYSIOLOGY

3 types of Influenza that infect humans: Types A, B, C.

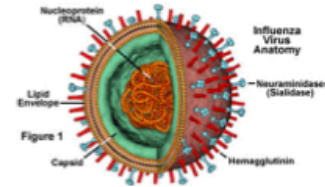
Types A&B are most important

-Surface Antigen (hemagglutinin & neuraminidase)-types HA and NA

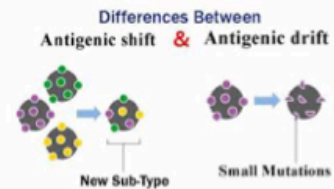
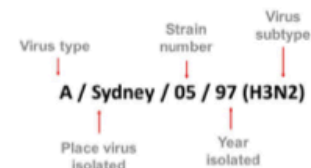
-Influenza evades the immune system by changing its antigenic properties

Antigenic Drift: small genetic changes that occur over time as the virus replicates. This is why annual flu vaccines are generated against the most prevalent evolving virus

Antigenic Shift: a major genetic change occurs resulting in new H or N proteins. This results in epidemics and pandemics



Understanding the naming of flu viruses



Antigenic shift is a major genetic change that creates a new HA and or NA combination, often through reassortment, and this large jump in antigenic profile can lead to severe epidemics or pandemics because the population has little preexisting immunity

INFLUENZA VACCINATION

Recommended ANNUALLY for all patients ages >6 mos

Inactivated

-killed vaccine given annually
-safe and effective in pregnancy, newborn has antibodies for 6 mos postpartum

Live Attenuated Nasal Spray

-approved for 2-49 yo
-observational studies show less effective in children than the inactivated vaccine



<https://www.cdc.gov/mmwr/volumes/72/rr/pdfs/rr7202a1-H.pdf>

Seasonal influenza vaccines are usually given as standard intramuscular "needle" shots, but needle free formulations also exist for patients who are needle averse

- Older adults receive a high dose influenza vaccine because immunosenescence makes them less likely to mount an adequate immune response to the standard dose, so providing a higher amount of antigen helps boost their antibody response and improves protection against flu

Immunosenescence = An age-associated phenomenon characterized by decreased function of the immune system as a result of multiple factors and results in increased susceptibility to infection, decreased response to vaccination, and increased risk of inflammatory diseases

INFLUENZA VACCINATION EFFICACY

Vaccine induced antibody to HA and NA

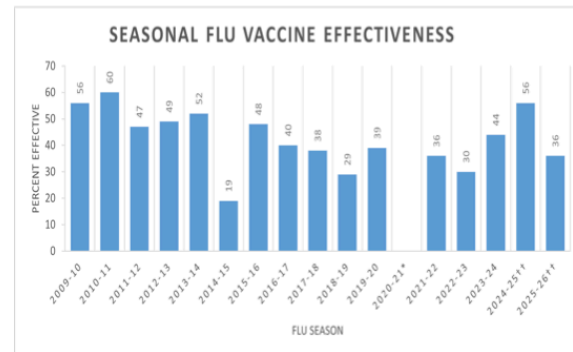
Antibodies develop in 10-14 days; last for 6-8 months

Good response in most vaccines

- Less response in immunocompromised
- Efficacy is higher in children

Efficacy affected by the match between vaccine antigen and epidemic virus

- Protection from illness, others have milder course
- Significant reduction in influenza deaths
- Much lower in years with poor match



CDC

COVID-19

Identified in late 2019 in Wuhan, China. An epidemic in China was followed by a global pandemic

In Feb 2020, WHO classified it as COVID-19, which stands for coronavirus disease 2019

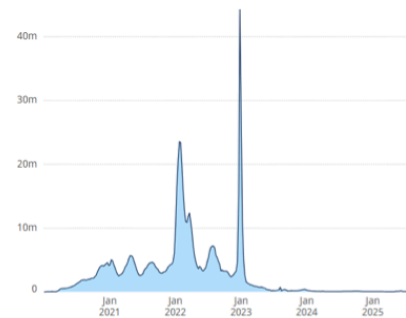
The virus that causes it is designated severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)

The WHO declared an end to the COVID-19 global health emergency in May 2023, more than three years after its emergence

Coronaviruses are enveloped positive-stranded RNA viruses

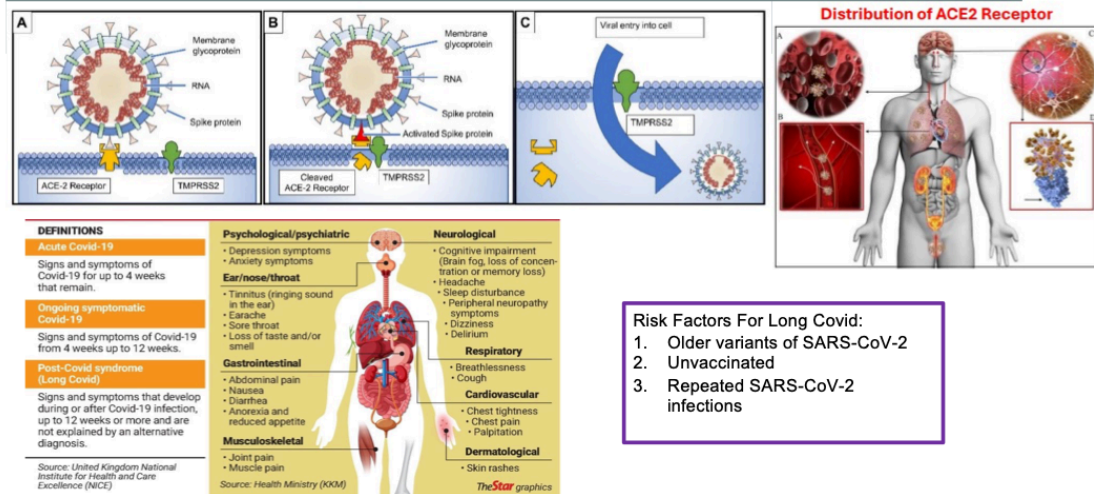
Total COVID-19 cases reported to WHO (weekly)

World, January 2020 - present



Source: World Health Organization

Pathogenesis of COVID-19



COVID-19 primarily affects organs that express the ACE2 receptor, because SARS-CoV-2 uses ACE2 as its entry point into cells, which explains why the lungs, cardiovascular system, GI tract, kidneys, and other ACE2 rich tissues can all be involved

COVID VACCINATION

Key updates to the 2025–2026 COVID-19 vaccination schedule

- For children ages 6 months–4 years, only Moderna (Spikevax) COVID-19 Vaccine is approved for use; Pfizer-BioNTech COVID-19 Vaccine is no longer authorized for this age group.
- Novavax COVID-19 Vaccine, approved for people ages 12 years and older, is administered as a single dose for initial vaccination.
- Moderna (mNexspike) COVID-19 Vaccine is approved for people ages 12 years and older.

Mechanism	Age
mRNA	≥6 mo
mRNA	≥ 6mos
Protein adjuvant	≥ 12 yr

Recommendations based on several factors, including,
(1) previous vaccine received (2) age (3) immune status
Recommended for age 6 months and greater
Recommend booster for this season

<https://www.cdc.gov/covid/hcp/vaccine-considerations/routine-guidance.html>

It's recommended for people to still be vaccinated against COVID even though it's not as deadly as the flu

COVID VACCINATION

COVID-19 vaccination, World data

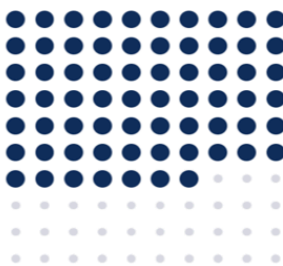
13.64bn
Total COVID-19 vaccine doses administered

Date of first COVID-19 vaccine product introduction
World

22 July 2020

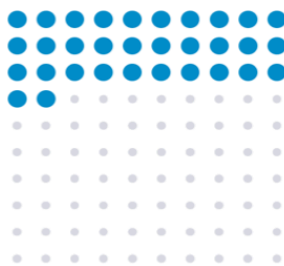
Percentage of total population vaccinated with a
complete primary series of a COVID-19 vaccine
World, 31 December 2023

67%



Percentage of total population vaccinated with at
least one booster dose of a COVID-19 vaccine
World, 31 December 2023

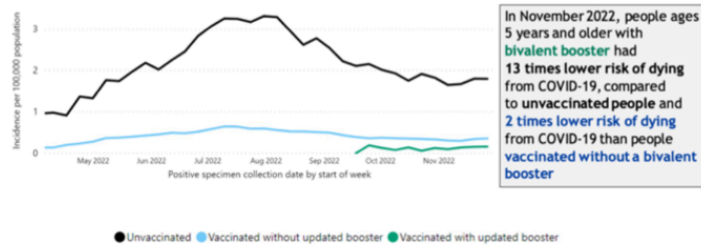
32%



Source: World Health Organization

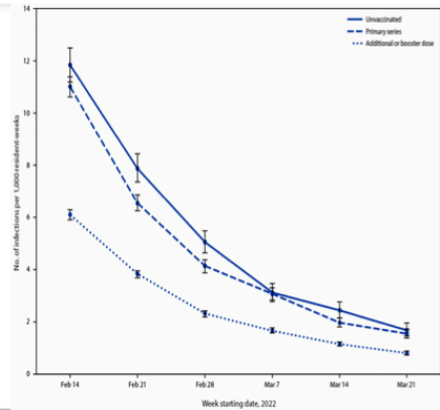
COVID VACCINATION: DO BOOSTERS WORK?

Death Rates by Vaccination Status and Receipt of Bivalent Booster Doses Among People Ages ≥12 Years April 3-December 3, 2022 (23 U.S. Jurisdictions)



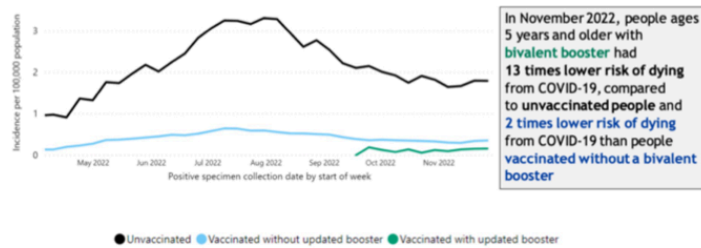
*Includes either a booster or additional dose.
<https://covid.cdc.gov/covid-data-tracker/#rates-by-vaccine-status>. Accessed Jan 25, 2023

CDC



COVID VACCINATION: DO BOOSTERS WORK?

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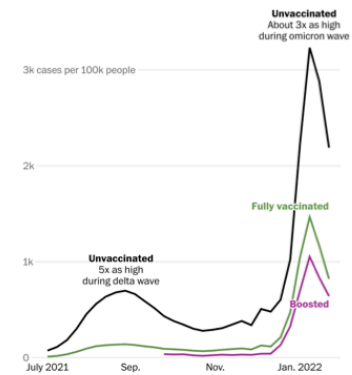


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<https://www.cdc.gov/covid-data-tracker/#rates-by-vaccine-status>. Accessed Jan 25, 2023

CDC

Cases by vaccination status

The spike of cases in the delta and omicron waves greatly reduced the difference in rates between vaccinated and unvaccinated.



RESPIRATORY SYNCYTIAL VIRUS

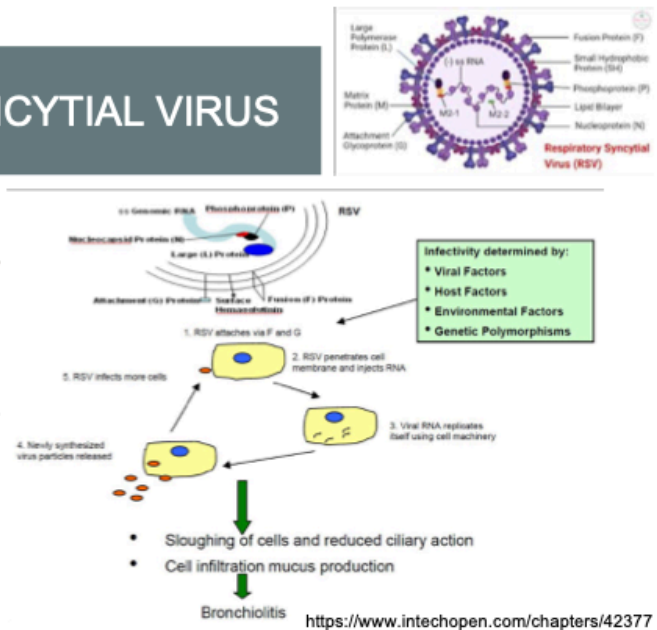
RSV is a single-stranded, negative-sense ribonucleic acid (RNA) virus

Two subtypes, A and B, are simultaneously present in most outbreaks, **with A subtypes typically causing more severe disease**

Typically causes **seasonal outbreaks**. In the northern hemisphere, these usually occur from October or November to April or May, with a **peak in January or February**

Transmitted primarily by direct contact to ocular or nasopharyngeal membranes.

Repeat infections common. After the 3rd infection, disease severity decreases.



Really important for extreme ranges of age groups

RESPIRATORY SYNCYTIAL VIRUS

Epidemiology

RSV causes acute respiratory tract illness in persons of all ages

CHILDREN:

It is the most common cause of lower respiratory tract infection (LRTI) in children <1 year of age

The global annual rate of RSV hospitalization among children <5 years was 4.4 per 1000

RSV hospitalization rates were highest among children <6 months (20.0 per 1000, 95% CI 0.7-41.3) and premature infants <1 year

ADULT:

Healthy Adults < 50y: repeat mild infections, mainly URT infections

Adults > 50y & those with underlying medical conditions:

-prospective studies, RSV was detected in ~ 10% of outpatients ≥60 years with acute respiratory illness

-the mortality rate among adults ≥50 years of age who are hospitalized with RSV is 6-8%

Pediatr Pulmonol. 2017;52(4):556. Epub 2016 Oct 14.
N Engl J Med. 2009;360(6):586.
Lancet. 2017;390(10098):946. Epub 2017 Jul 7.
PLoS One. 2017;12(8):e0182321. Epub 2017 Aug 10.

RESPIRATORY SYNCYTIAL VIRUS

Presenting Symptoms

RSV (Respiratory Syncytial Virus)

Symptoms of RSV can affect
age groups differently.

Symptoms in babies:



- Fussiness or irritability.
- Decreased appetite or difficulty feeding.
- Minimal interest in activities.
- Changes to their breathing pattern.
- Fever.
- Cough.

Symptoms in toddlers:



- Runny nose.
- Coughing and sneezing.
- Difficulty eating or drinking.
- Little interest in playing.
- Trouble swallowing.
- Breathing faster or slower than normal.
- Fever.

Severe symptoms in babies and toddlers:

If your baby or toddler has any of the following
symptoms, visit the emergency room or call 911
immediately:

- Noisy breathing.
- Flaring of nostrils while breathing.
- Blue or gray color to their lips, mouth and fingernails.
- Belly breathing or caving in of the chest with expansion of the belly when breathing.
- Short, shallow or rapid breathing.
- Pauses while breathing.

Symptoms in children and adults:



- Runny nose.
- Congestion.
- Mild headache.
- Sore throat.
- Cough.
- Fatigue.
- Fever.

Cleveland Clinic

RESPIRATORY SYNCYTIAL VIRUS Vaccination

3 RSV vaccines available
-all are approved for adults 50+
-only ABRYYSVO is approved for
pregnant patients

Protein subunit vaccines
GSK's AREXVY
Pfizer's ABRYYSVO

mRNA vaccine
Moderna's mRESVIA

Who Does It Protect?	Type of Product	Who Is It Recommended For?	When Is It Available?
 Adults 50 and over	RSV vaccine	Adults ages 50-74 who are at increased risk of severe RSV AND Everyone ages 75 and older	Available any time, but best time to get vaccinated is late summer and early fall
 Babies	RSV antibody given to baby	All infants whose mother did not receive RSV vaccine during pregnancy, and some children ages 8-19 months who are at increased risk for severe RSV	October through March*
 Babies	OR RSV vaccine (Pfizer's ABRYYSVO) given to mother during pregnancy	All pregnant women during weeks 32-36 of their pregnancy	September through January

www.cdc.gov/rsv

*Recommended timing of administration in most of the continental United States. Recommended timing of administration may differ in some areas, based on state, local, or territorial guidance.



RESPIRATORY SYNCYTIAL VIRUS

Vaccination for ages 50-74

Factors associated with increased risk* for severe RSV disease include:



Chronic lung disease



Chronic cardiovascular disease



End-stage renal disease or dependence on dialysis



Diabetes mellitus with end-organ damage or requiring insulin or SGLT2 inhibitor



Moderate or severe immunocompromise



Chronic or progressive neurological or neuromuscular conditions

Other factors include:

- » Chronic liver disease
- » Chronic hematologic conditions
- » Severe obesity (BMI ≥ 40 kg/m²)
- » Residence in a nursing home
- » Other conditions or factors that put your patient at increased RSV disease risk

*Self-attestation is sufficient evidence of a risk factor.

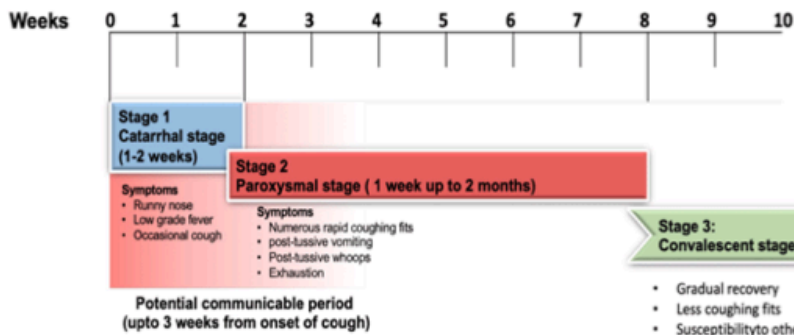
Only 1 vaccine is needed in adults

Vaccination reduces the risk for RSV Hospitalizations by 75%

2025-2026 Combined Respiratory Vaccine Recommendations			
	Influenza Vaccine	RSV Immunization	COVID-19 Vaccine
Infants and Children	All children 6 months and older Some children 6 months to 8 years may need multiple doses AAP, CDC	All infants <8 months + children 8-19 months with risk factors (nirsevimab, clesrovimab), typically Oct-Mar, if no maternal RSV vaccine AAP, CDC	All children 6-23 months + children 2-17 years with risk factors or if parent desires vaccine AAP
Pregnancy	All At any point in pregnancy ACOG, CDC	32-36 weeks gestation (Pfizer, Abrysvo only) Typically Sept-Jan ACOG, CDC	All At any point in pregnancy ACOG
Adults 18-50	All CDC	See Pregnancy CDC	All* *2024-2025 medical society recommendations remain until updated (emphasis on high risk adults) ACIP, AAP, IDSA
Adults 50+	All High-dose, recombinant or adjuvanted flu vaccine preferred for 65+, if available CDC	All 75+ and adults 50-74 with risk factors One lifetime dose of RSV vaccine CDC	All* *2024-2025 medical society recommendations remain until updated (emphasis on high risk adults) ACIP, AAP, IDSA
 COMMON HEALTH Coalition Together for better health Last updated 9/2/25			

PERTUSSIS (Whooping Cough)

Disease progression of pertussis



Pertussis or whooping cough is vaccine preventable and current recommendations include vaccination for all pregnant women and for close contacts such as spouses and family members to create a protective cocoon around the newborn

- The illness progresses through stages, beginning with a catarrhal phase where symptoms resemble a mild URI with cough and runny nose
- This is followed by a paroxysmal phase that can last weeks to months, characterized by repeated, forceful coughing fits that may be so intense they cause posttussive vomiting and severe exhaustion
- The classic inspiratory “whoop” occurs at the end of a coughing bout when the patient finally manages to take a breath in through narrowed airways
- After this prolonged phase, patients slowly enter a convalescent stage with gradual reduction in cough frequency, but they may look back on months of relentless coughing that was never recognized as pertussis
- Early diagnosis allows relatively simple treatment, yet the key clinical and public health message is that disease is far more easily prevented with appropriate vaccination

PERTUSSIS VACCINATION

People of all ages need
WHOOPIING COUGH
VACCINES

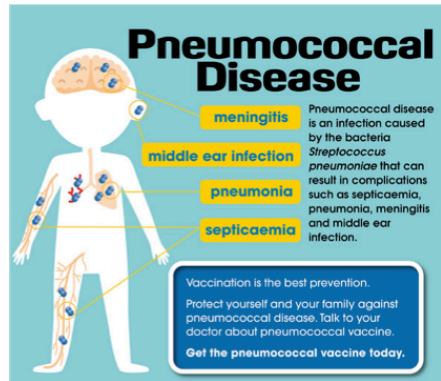


DTaP for young children	Tdap for preteens	Tdap for pregnant women	Tdap for adults
✓ 2, 4, and 6 months ✓ 15 through 18 months ✓ 4 through 6 years	✓ 11 through 12 years	✓ During the 27-36th week of each pregnancy	✓ Anytime for those who have never received it

www.cdc.gov/whoopingcough



PNEUMOCOCCAL VACCINATION



Adults 65 years of age and older & adults living with a **chronic lung condition** are among those at **highest risk** of pneumococcal infection.

Immunization against pneumococcal disease is recommended for people with the following chronic lung conditions:

ASTHMA

COPD

CYSTIC
FIBROSIS

PULMONARY
HYPERTENSION

PNEUMOCOCCAL VACCINATION

Types of Pneumococcal Vaccination

- Pneumococcal conjugate vaccines (PCVs, specifically PCV15 and PCV20)
- Pneumococcal polysaccharide vaccine (PPSV23)

Candidates

- Routine Vaccination for Infants and Children
 - Give PCV15 or PCV20 to infants as a series of 4 doses, one dose at each of these ages: 2 months, 4 months, 6 months, and 12 through 15 months.
- Adults 65 years & older without an immunocompromising condition, cerebrospinal fluid leak or cochlear implant
- Adults 19 through 65 years with certain high-risk conditions

"These are not things that are memorized. These are things that you'll learn through your rotations."

Case Studies

Case 1

23-year-old

ACUTE INFLUENZA INFECTION

Yearly Influenza Vaccine

Up to Date COVID
Vaccination

Case 2

58-year-old

IDIOPATHIC PULMONARY FIBROSIS

Yearly Influenza Vaccine

Up to Date COVID
Vaccination

Pneumococcal Vaccination
Anticipatory Guidance that
at 60 years should receive
RSV vaccine

Case 3

13-year-old

PERSISTENT POORLY CONTROLLED ASTHMA

Yearly Influenza Vaccine

Up to Date COVID
Vaccination

Be sure received Tdap
between 11-12 years

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



Technique & Interpretation of the lung and thorax PE



Basic pulmonary diagnostic techniques



Understand High Risk behavior for lung disease



Review Vaccine Preventable Respiratory Illness



Review additional Preventable Diseases

TREATMENT OF LATENT TUBERCULOSIS

Latent TB is now termed Tuberculosis Infection (TBI)

Active TB is now termed Tuberculosis Disease

TBI

Infected by *Mycobacterium tuberculosis* and the host immune system contains the organism

Asymptomatic and noninfectious but harbor potentially viable organisms

Organisms have the potential to overcome host immune defenses and subsequently cause symptomatic disease (reactivation TB)

The treatment of TBI kills the contained organisms, decreasing the risk of developing TB disease by 90%

Latent TB is now termed tuberculosis infection (TBI) and is distinct from active tuberculosis disease

- TBI means the patient has been infected with *Mycobacterium tuberculosis*, the immune system has contained the organisms, and the person is asymptomatic and noninfectious, yet viable bacilli remain and can reactivate later
- Treating TBI is crucial because it kills these contained organisms and decreases the risk of progressing to active TB disease by about 90 percent, which protects both the patient and their contacts
- Immunocompromised patients are a top priority for testing and treatment of latent TB

TBI

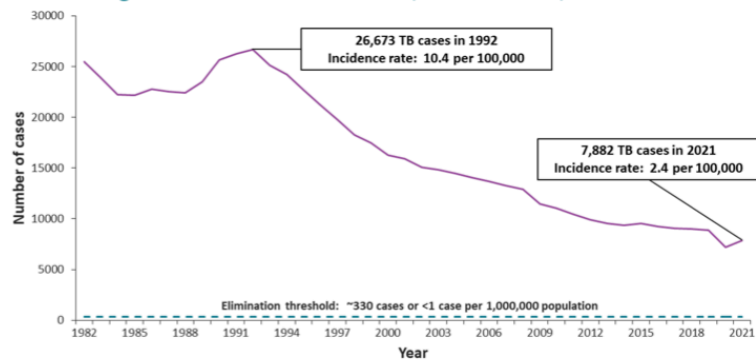
For people with TB infection, no risk factors, and no treatment, the risk is about 5% in the first 2 years after infection and about 10% over a lifetime.

About half of those people who develop TB Disease will do so within the first two years of infection.

For people with untreated TB infection and untreated HIV infection, the risk is about **7% to 10% PER YEAR**.

TBI

Progress Towards TB Elimination, United States, 1982–2021



Not So Fast...

Of concern, TB cases in the US have been increasing since the covid pandemic

Nationwide, 2023 saw the highest number of cases reported in the last decade, a 16% increase from 2022 and an 8% increase from 2019.

Preliminary data from the CDC indicates > 10,300 cases in 2024, an 8% increase from 2023

Most of these recent cases are not instances of novel transmission, but **activation of latent infections that were undetected or improperly treated.**

This is thought due to public health resource diversion



Increase in TB cases in the United States, according to CDC surveillance:



Healio

Better Control in 2025!

10,260 cases in 2025 vs 10,395 in 2024

Non-U.S.-born persons accounted for 7,858 cases (77% of the total),
with an incidence rate of 15.4 per 100,000 persons,
whereas U.S.-born persons accounted for 2,252 cases (22%)
at a rate of 0.8 per 100,000.



<https://www.cdc.gov/tb-data/aboutprovisionaldata/2025-provisional-data.html>

TBI

2 types of tests can detect TB bacteria in the body. The tests DO NOT decipher between TBI and TB disease

TBI is asymptomatic with a normal CXR but a positive TB test

1. TB Skin Test (TST)
2. TB Blood Test

High Risk Patients: HIV, IVDA, ESRD, DM, Silicosis, Immunosuppressive therapy, hematologic cancers, Malnourished (loss of > 10% BW), Gastrectomy

Screen these individuals for tuberculosis infection⁴⁻⁶

Those at high risk of acquiring latent infection

- Household contacts and others in close contact with an individual who has active tuberculosis
- Mycobacteriology laboratory personnel
- Immigrants from countries with high rates of tuberculosis*
- Residents and staff of high-risk congregate settings (homeless shelters and correctional facilities)
- Populations determined to be at high risk by local or state health departments

Infected individuals at high risk of progressing to active disease

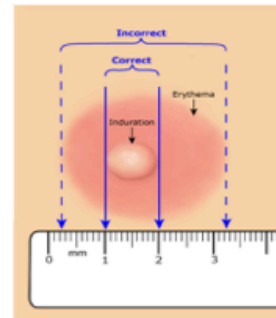
- Patients infected with HIV or another condition that weakens the immune system (eg, silicosis, diabetes mellitus)
- Patients receiving immunosuppressive therapy (including chemotherapy and tumor necrosis factor- α inhibitors) and recipients of organ transplants
- Children <5 years
- Those with chest x-ray findings consistent with old tuberculosis
- Users of injectable illicit drugs

Other high risk patients include close household contacts of active TB cases, mycobacteriology lab workers, immigrants from high prevalence countries, residents and staff of homeless shelters or correctional facilities, and young children exposed to high risk adults

Mantoux Tuberculin Skin Test



The tuberculin material stimulates a **delayed-type hypersensitivity response mediated by T lymphocytes** and, in patients with prior mycobacterial exposure, causes induration at the injection site within 48 to 72 hours.



Only the raised, pencil eraser like induration is measured for the skin test, not the surrounding redness

Mantoux Tuberculin Skin Test



1

ADMINISTRATION

2

READING

3

INTERPRETATION

Induration of ≥ 5 mm is considered positive in

- People living with HIV
- Recent contacts of people with infectious TB disease
- People who have fibrotic changes on a chest radiograph
- Patients with organ transplants
- Other immunosuppressed patients (e.g., patients on prolonged therapy with corticosteroids ≥ 15 mg per day of prednisone or those taking TNF- α antagonists)

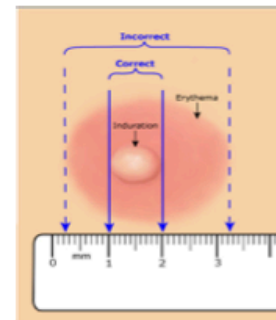
Induration of ≥ 10 mm is considered positive in

- People born in countries where TB disease is common, including Mexico, the Philippines, Vietnam, India, China, Haiti, and Guatemala
- People who misuse drugs and alcohol
- People who live or work in high-risk congregate settings (e.g., nursing homes, homeless shelters, or correctional facilities)*
- Mycobacteriology laboratory workers
- People with certain medical conditions that place them at high risk for TB (e.g., silicosis, diabetes mellitus, severe kidney disease, certain types of cancer, or certain intestinal conditions)
- Children younger than 5 years of age
- Infants, children, and adolescents exposed to adults in high-risk categories

Induration of ≥ 15 mm is considered positive in

- People with no known risk factors for TB

Positive Induration Size Depends on Risk Factors



Interpretation depends on both size and risk, with:

- ≥ 5 mm considered positive in very high risk patients like people with HIV, recent close contacts, fibrotic changes on chest x ray, organ transplants, or other major immunosuppression
- ≥ 10 mm positive in lower but still high risk patients such as immigrants from high prevalence countries such as Mexico, the Philippines, Vietnam, India, China, Haiti, and Guatemala, people who misuse drugs, residents or workers in high risk congregate settings, mycobacteriology lab personnel, patients with high risk medical conditions, children < 5 years, and children exposed to high risk adults
- ≥ 15 mm positive in people with no known risk factors

MANTOUX TUBERCULIN SKIN TEST

FALSE NEGATIVE

Immunosuppression
Too recent of an infection
Improper
handling/interpretation

FALSE POSITIVE

Nontuberculous
mycobacteria infection
Bacillus Calmette-Guérin
(BCG) vaccination

Many centers are shifting toward blood based tests that are less subjective and are not eyeballed

TB BLOOD TEST

Interferon Gamma Release Assays (IGRA)

-QuantIFERON®-TB Gold Plus (QFT-Plus)

-T-SPOT®.TB test (T-Spot)



In vitro blood tests of **cell-mediated immune response**; ELISA detects T cell release of interferon-gamma following stimulation by 2 antigens unique to *Mycobacterium tuberculosis* and a few other mycobacteria (*M kansasii*, *M szulgai*, *M marinum*)

Result: Positive, Negative, Indeterminate (repeat testing in 8 wks)

Preferred use: those with prior BCG vaccine, those with compliance concern

Limited data in children < 5 years.

Interferon gamma release assays such as QuantiFERON are blood tests for TB infection that are preferred over the tuberculin skin test in many settings because they are more objective and less affected by prior BCG vaccination or exposure to some nontuberculous mycobacteria, which can confound skin test

- These tests measure T cell release of interferon gamma in vitro after stimulation with TB specific antigens and report a simple result of positive, negative, or indeterminate
- They are "one and done" because the patient only needs a single blood draw and does not have to return 48–72 hours later for reading, which reduces loss to follow up and improves screening in populations with compliance concerns
- An indeterminate result means the assay controls did not behave as expected

TREATEMENT OF TBI

All patients with TBI should be screened for HIV

4 Regimens:

MEDS	FREQUENCY	DURATION
INH + Rifapentine	Weekly	3 months
Rifampin	Daily	4 months
INH + Rifampin	Daily	3 months
INH	Daily	6-9 months <i>(not preferred due to SE's & lower completion rate)</i>

INH not used as often due to toxicity

VENOUS THROMBOEMBOLISM

The most common presentations of venous thrombosis:

- Deep vein thrombosis (DVT) of the lower extremities
- Pulmonary embolism

Inherited Risk Factors

Factor V Leiden mutation
Prothrombin G20210A mutation
Antithrombin deficiency
Protein C deficiency
Protein S deficiency



Acquired Risk Factors

>60 yo	Trauma
Obesity	Pregnancy
Malignancy	central line
Prior DVT/PE	CHF
Immobilization	CHD
Surgery, esp Ortho	Nephrotic Syndrome
Anesthesia > 30 min	Antiphospholipid syndrome
Estrogen therapy	Covid-19
Stroke	
Sepsis	

Prevention of pulmonary embolism starts with preventing venous thrombosis, most commonly DVTs of the lower extremities, which then embolize to the lungs

- Hospitalized patients who are immobilized and not given appropriate DVT prophylaxis and patients with active cancer are especially high risk in clinical practice

PULMONARY EMBOLISM

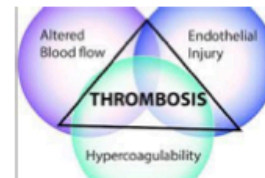
Complication of DVT

90% of emboli arise from the deep veins of the lower extremities

Risk of emboli depends on the location—above or below the knee

AKDVT present the greatest embolic risk

AKDVT = Above the Knee DVT



On CT pulmonary angiography above, a massive “saddle” PE appears as thrombus straddling the bifurcation of the main pulmonary artery and extending into both right and left main branches, which can critically obstruct pulmonary blood flow and rapidly lead to hemodynamic collapse and death

PREVENTION OF VENOUS THROMBOEMBOLISM



Preventing the formation of thrombi is the primary method to prevent fatal PE

Without treatment, death is usually due to recurrent emboli w/in hours of the initial embolic event

Many embolic episodes with survival beyond a few hours are never diagnosed

Most deaths, therefore, can be prevented by preventing the embolic event

Blood Clots are Deadly and a Significant, Growing Public Health Problem



Although there are many reasons a person might develop a blood clot, about half of them are directly related to a recent hospitalization or surgery and most of these do not occur until after discharge.

Venous thromboembolism prevention is a core inpatient responsibility, so you must decide which admitted patients need DVT prophylaxis and which can safely go without it

- Low risk patients having brief ambulatory procedures who are up and walking the same day generally do not require pharmacologic prophylaxis, whereas patients admitted with acute medical illnesses such as COPD with pneumonia or those undergoing major orthopedic procedures are high risk and must receive DVT prophylaxis
- Roughly half of all VTE events are health care associated, which means preventable failures in hospital prophylaxis can lead to fatal pulmonary emboli
- Mechanical strategies include early mobilization and use of intermittent pneumatic compression devices on the legs, which cyclically squeeze the calves like a blood pressure cuff to mimic muscle contraction and promote venous return

PREVENTION OF VENOUS THROMBOEMBOLISM

Early postop mobility

Intermittent pneumatic compression device

Low dose unfractionated heparin (drip, sq)

Low molecular weight heparins (enoxaparin)

Oral anticoagulants



VTE prevention strategies should be individualized according to the risk of VTE (very low, low, moderate, and high) as well as the risk and consequences of major bleeding

VTE prophylaxis should be continued until the patient becomes fully ambulatory or until hospital discharge

Extended pharmacologic VTE prophylaxis beyond discharge is not routinely recommended except for patients who undergo major abdominal and/or pelvic surgery for cancer in whom thromboprophylaxis should be continued for a period of four weeks & in major orthopedic surgeries where VTE prophylaxis is extended up to 35 days

Some patients only need prophylaxis while hospitalized, but those undergoing high risk surgeries such as joint replacement or major abdominal and pelvic cancer operations often require extended prophylaxis for weeks after discharge to prevent late postoperative DVT and PE

Case Studies

Case 1

23-year-old

**ACUTE INFLUENZA
INFECTION**

Case 2

58-year-old

**IDIOPATHIC
PULMONARY FIBROSIS**

Case 3

13-year-old

**PERSISTENT POORLY
CONTROLLED ASTHMA**

Consider PPD to r/o TBI based on Risk Factors

Consider PPD to r/o TBI
if need immune
modulating medications

OVERVIEW



Taking a medical history of patients with lung disease



Pathophysiology of Presenting Symptoms



Technique & Interpretation of the lung and thorax PE



Basic pulmonary diagnostic techniques



Understand High Risk behavior for lung disease



Review Vaccine Preventable Respiratory Illness



Review additional Preventable Diseases



THANK YOU!